

Q5: Evaluation Metrics from a Multi-Class Confusion Matrix

The system classified 90 animals into Cat, Dog, or Rabbit. The results are shown below:

System \ Gold	C at	Do g	Rabb it
Cat	5	10	5
Dog	15	20	10
Rabbit	0	15	10

Tasks:

1. Per-Class Metrics

- Compute precision and recall for each class (Cat, Dog, Rabbit).

Cat

- $TP = 5$
- Predicted Cat = 20 $\rightarrow$ $FP = 15$
- Actual Cat = 5 $\rightarrow$ $FN = 15$

Precision = $5 / 20 = \mathbf{0.25}$

Recall = $5 / 20 = \mathbf{0.25}$

Dog

- $TP = 20$
- Predicted Dog = 45 $\rightarrow$ $FP = 25$
- Actual Dog = 20 $\rightarrow$ $FN = 25$

Precision = $20 / 45 \approx \mathbf{0.444}$

Recall = $20 / 45 \approx \mathbf{0.444}$

Rabbit

- $TP = 10$
- Predicted Rabbit = 25 $\rightarrow$ $FP = 15$
- Actual Rabbit = 10 $\rightarrow$ $FN = 15$

Precision = $10 / 25 = \mathbf{0.40}$

Recall = $10 / 25 = \mathbf{0.40}$

2. Macro vs. Micro Averaging

- Compute the macro-averaged precision and recall.

Macro-averaged precision & recall

Macro = average of per-class values

- Macro Precision = $(0.25 + 0.444 + 0.40) / 3 \approx \mathbf{0.365}$
- Macro Recall = $(0.25 + 0.444 + 0.40) / 3 \approx \mathbf{0.365}$

- Compute the micro-averaged precision and recall.

Micro-averaged precision & recall

Micro = compute globally across all classes

- Total TP = $5 + 20 + 10 = 35$
- Total Predicted = 90
- Total Actual = 90

Micro Precision = $35 / 90 \approx \mathbf{0.389}$

Micro Recall = $35 / 90 \approx \mathbf{0.389}$

- Briefly explain the difference in interpretation between macro and micro averaging.

Macro averaging: treats all classes equally → good when classes are imbalanced, shows average per-class performance.

Micro averaging: weights by the number of samples per class → better reflects overall system performance across all instances.

3. Programming Implementation

Write Python code that:

1. Accepts the confusion matrix above as input.
2. Computes per-class precision and recall.
3. Computes macro-averaged and micro-averaged precision and recall.
4. Prints all results clearly.